

VARSHA SATHISKUMAR

Denver, CO | (720)-255-6441 | varsha_sathiskumar@mines.edu | [linkedin.com/in/varsha-sathiskumar](https://www.linkedin.com/in/varsha-sathiskumar) | varshasathiskumar.github.io

PROFESSIONAL SUMMARY

Software Engineer with 4+ years of experience building scalable full-stack enterprise and B2B web applications, with recent experience developing Agentic AI systems using RAG, LLMs, and multi-agent architectures. Experienced in taking features from design to production and delivering measurable business impact.

EDUCATION

Colorado School of Mines

Master of Science in Computer Science - GPA: 3.6/4.0

| Golden, CO

Dec 2026

Thiagarajar College of Engineering

Bachelor of Technology in Information Technology - GPA: 8.7/10.0

| Madurai, India

Apr 2021

WORK EXPERIENCE

AI Software Engineer Intern | NociQuant Inc, Minneapolis, MN

Jun 2026 – Aug 2026

- Developed a full-stack healthcare application with Python backend services integrating an AI diagnostic pipeline and a React interface for reviewing diagnostic results, supporting earlier chronic pain diagnosis.
- Built an end-to-end diagnostic pipeline integrating RAG, multi-agent orchestration, and an automated evaluation framework, improving retrieval recall by 20% across 1,700+ research papers and diagnostic accuracy from 68% to 85% across 1,000 clinical cases using Python, LangChain, and LangGraph.
- Automated evaluation framework to validate 6 AI diagnostic configurations across 20+ evaluation criteria, identifying accuracy and retrieval failures through Python-based test automation and regression analysis.

Full Stack Software Engineer | Kimberly Clark, India

May 2023 – Mar 2025

- Owned full-stack development of an enterprise pricing application, delivering 50+ features across 4 global regions and processing 200+ pricing agreements weekly, achieving over 97% user acceptance using C# and ASP.NET Core.
- Led the implementation of 10+ REST APIs integrating SAP and Salesforce, supporting 15K+ monthly API transactions for the pricing workflow using Node.js and Redis.
- Migrated a legacy C# .NET codebase to ASP.NET Core and modernized it into 4 independently deployable microservices, improving service isolation and reducing production support incidents by 32% using Docker and Kubernetes.
- Led the migration of 10+ legacy SSIS packages to Azure Data Factory, reducing batch processing from 4 hours to 2.5 hours and making business-critical data available 1.5 hours earlier for downstream reporting and analytics.
- Eliminated timeout failures and reduced 10K-record report generation from 3–4 minutes to under 10 seconds across 4 regional workflows by replacing cursor-based processing with set-based SQL.

Associate Software Engineer | Kimberly Clark, India

Sep 2021 – Apr 2023

- Developed Onvation, an IoT smart-facility platform enabling real-time restroom monitoring across 15+ U.S. states and Canada, contributing to 105% higher visitor satisfaction and 80% lower consumable waste using Python, React, and JavaScript.
- Built serverless Azure pipelines processing 1M+ daily IoT events for threshold monitoring and automated alerts, reducing 15+ monthly service requests and saving \$68,052 annually per user using Azure IoT Hub, Cosmos DB, and Azure Functions.
- Optimized Azure CI/CD pipelines across 4 repositories, reducing deployment time from 25 to 12 minutes and enabling faster, more frequent releases through Docker caching, parallel execution, and automated quality gates.
- Expanded automated test coverage from 1,200 to 1,800+ test cases across 100+ workflows, catching defects and regressions before release using Jest, pytest, and Playwright.

Software Engineer Intern | Kimberly Clark, India

Jan 2021 – Aug 2021

- Designed and deployed a centralized consulting request platform serving 3,000+ global team members and processing 75+ weekly requests, reducing processing time from 1 day to 4 hours using Power Apps and Power Automate.
- Built 15+ reusable UI components adopted across 10+ teams, improving application UI consistency and reducing duplicated development effort using Microsoft Power Apps, Power Fx, and JavaScript.

PROJECTS

Geospatial Web Platform | GitHub

| Jan 2026 – May 2026

- Engineered a GIS platform with 12 interactive map layers for Delaware Mountain Group exploration, reducing geological data query time from over 40 minutes to ~30 seconds using React, FastAPI, and Mapbox GL JS.
- Developed an ETL pipeline consolidating 2TB+ of geological data from 30+ source files and processing 10GB+ per execution, eliminating manual consolidation and standardizing geospatial data storage using PostgreSQL, Supabase, and PostGIS.

- Implemented role-based access control across 22 backend APIs and 3 permission levels, achieving an 85% pass rate across 350 security and validation test cases using OIDC, OAuth 2.0, and JWT.

Deep Research Multi-Agent System | [GitHub](#)

| Sep 2026

- Built a 4-agent research system to automate document analysis, paper discovery, research-gap identification, and report synthesis by orchestrating specialized agents through LangGraph and OpenAI APIs.
- Engineered reliable agent workflows using structured outputs, schema validation, and retry mechanisms, equipping agents with vector-search and web-search tools for semantic document retrieval and research-paper discovery.

Personal Portfolio with AI Voice Agent | [GitHub](#)

| Sep 2026

- Built a RAG-powered AI voice agent, enabling conversational Q&A grounded in portfolio and resume content, by integrating OpenAI APIs with vector-based retrieval.
- Automated portfolio website development, Playwright-based web testing, and code-review workflows by orchestrating specialized coding, reviewer, and conversational agents using the Claude Agent SDK.

Product Review Summarization Platform | [GitHub](#)

| May 2026

- Built a distributed NLP pipeline processing 22GB of Amazon reviews, achieving 84.03% sentiment accuracy and 0.8431 weighted F1 while generating product-level intelligence on sentiment, strengths, weaknesses, and recurring complaints using Llama 3.1, BGE-M3, and BERTopic.
- Implemented Power BI dashboards analyzing review data across 12 product categories, visualizing sentiment trends, review activity, complaint rates, and semantic clusters to support product buyers and business stakeholder analysis.

Assessing Response Consistency across Query Variants in LLM | [GitHub](#)

| Dec 2025

- Built an automated benchmarking pipeline comparing 5 OpenAI models across 10,000+ paraphrased queries, reducing processing from 3,199 to 1,530 seconds, enabling faster model and prompt-consistency analysis using Python, DuckDB, and Apache Arrow UDFs.
- Benchmarked Chain-of-Thought prompting, improving technical-query accuracy by 15% versus 3% on general-text queries, across 2,000 paraphrased queries and 5 OpenAI models using ground-truth validation.

TECHNICAL SKILLS

Programming Languages & Frameworks: Python, C#, SQL, JavaScript, TypeScript, HTML, CSS, React, Node.js, FastAPI, Jest, pytest, Playwright, Selenium, Cypress

Development Tools & Platforms: PostgreSQL, Microsoft SQL Server, MySQL, Supabase, Azure Cosmos DB, Redis, AWS, Microsoft Azure, Docker, Kubernetes, Azure DevOps, Git, GitHub, Jira, Postman, Swagger/OpenAPI

AI/ML Tools: TensorFlow, Keras, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, Hugging Face, Sentence-Transformers, LangChain, LangGraph, MCP, Jupyter Notebook, Pinecone, Claude Code